

Set : $A = \{1, 2\}$; $B = \{1, 2, 2\}$; $C = \{1, 2, 3\}$

↓
 $(G, \oplus)$

A and B are same sets

Binary Operator : If G is a non-empty set and $\otimes$ is defined on G then $\otimes$ is said to be a binary operator iff:-

$\otimes : G \times G \rightarrow G$: This means that cross maps $G \times G$ to G .

Example: $G \equiv \mathbb{R}$ & $\otimes \equiv +$, Then $+$ is a binary operator

Properties :-

- commutative : $a \otimes b = b \otimes a \quad \forall a, b \in G$

Eg:-

$G = \mathbb{R}$; $\otimes = +$

$G = \mathbb{R}$; $\otimes = \times$

$G = \text{Set of matrices}$; $\otimes = \times$ (Not commutative)

- associative : $a \otimes (b \otimes c) = (a \otimes b) \otimes c$

- distributive : $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$ (left distributive)

$(a \otimes b) \odot c = (a \otimes c) \odot (b \otimes c)$ (right distributive)

- Identity element :- An element $e \in G$ is said to be left identity element if:

$$e \otimes a = a \quad \forall e, a \in G$$

- Inverse of an element : b will be the inverse of a iff:

$$\begin{aligned} b \otimes a &= e && \text{(left inverse)} \\ a \otimes b &= e && \text{(right inverse)} \end{aligned}$$

Group

A group is an algebraic structure $(G, \otimes)$ which satisfies the following:-

$$\textcircled{1} \quad a \otimes (b \otimes c) = (a \otimes b) \otimes c \quad \forall a, b, c \in G$$

$$\textcircled{2} \quad \forall a \in G \exists e \quad \text{s.t.} \quad a \otimes e = e \otimes a = a$$

$$\textcircled{3} \quad \forall a \in G \exists b \quad \text{s.t.} \quad a \otimes b = b \otimes a = e$$

(Q.) If G is a group s.t. $(ab)^2 = a^2b^2 \quad \forall a, b \in G$ then prove that G is Abelian group (commutative).

$$(ab)^2 = a^2b^2$$

$$(a \otimes b) \otimes (a \otimes b) = (a \otimes a) \otimes (b \otimes b)$$

Let $c = a \otimes b \in G$,

$$\text{LHS: } c \otimes (a \otimes b) \Rightarrow (c \otimes a) \otimes b$$

$$[(a \otimes b) \otimes a] \otimes b \Rightarrow [a \otimes (b \otimes a)] \otimes b$$

Assuming $b \otimes a = a \otimes b$

$$[a \otimes (b \otimes a)] \otimes b = [a \otimes (a \otimes b)] \otimes b$$

$$= [(a \otimes a) \otimes b] \otimes b$$

$$= (d \otimes b) \otimes b, \quad d \in G, d = a \otimes a$$

$$= d \otimes (b \otimes b)$$

$$= (a \otimes a) \otimes (b \otimes b) = \text{RHS}$$

Hence our assumption is correct. G is an Abelian Group.

Groups, Rings & Fields

(Q.) Verify whether a set of $n \times n$ non singular matrices with real elements forms a group under matrix addition

Let $A_{n \times n}$ be a non-singular matrix with real elements $(G, +)$

$$G = \left\{ \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1n} \\ r_{21} & r_{22} & & r_{2n} \\ \vdots & \vdots & & \vdots \\ r_{n1} & r_{n2} & & r_{nn} \end{bmatrix} : r_{ij} \in \mathbb{R}, n \in \mathbb{N} \right\}$$

$$\det(G) \neq 0$$

- Test of Associativity : considers $A, B, C \in G$

$$A + (B + C) = \begin{bmatrix} a_{11} + (b_{11} + c_{11}) & \dots & a_{1n} + (b_{1n} + c_{1n}) \\ a_{21} + (b_{21} + c_{21}) & & a_{2n} + (b_{2n} + c_{2n}) \\ \vdots & & \vdots \\ \dots & & \dots \end{bmatrix}$$

$$= \begin{bmatrix} (a_{11} + b_{11}) + c_{11} & \dots & (a_{1n} + b_{1n}) + c_{1n} \\ (a_{21} + b_{21}) + c_{21} & \dots & (a_{2n} + b_{2n}) + c_{2n} \\ \vdots & \ddots & \vdots \end{bmatrix}$$

$$= (A + B) + C$$

- Test of Identity :- consider $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix} \in G$

Let $E = \begin{bmatrix} e_{11} & \dots & e_{1n} \\ e_{21} & \dots & e_{2n} \\ \vdots & \ddots & \vdots \end{bmatrix} \in G$ if possible $(a_{ij}, e_{ij} \in \mathbb{R})$

If $A + E = A$, then ; $\begin{bmatrix} a_{11} + e_{11} & a_{12} + e_{12} & \dots & a_{1n} + e_{1n} \\ a_{21} + e_{21} & a_{22} + e_{22} & \dots & a_{2n} + e_{2n} \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix}$

For two matrices to be equal, their corresponding elements must be equal
 $\Rightarrow$ for i, j^{th} element

$$a_{ij} + e_{ij} = a_{ij} \Rightarrow e_{ij} = 0$$

$$\Rightarrow E = \begin{bmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}_{n \times n} \text{ (Null MATRIX)} \Rightarrow \det(E) = 0 \Rightarrow E \notin G$$

This contradicts our assumption. Identity element does not exist
 $\Rightarrow (G, +)$ is not a group.

modification: Let G now be the set of all $n \times n$ matrices whether singular or not. Then, identity element E exists, $E \in G$.

- Test for inverse

Let $A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & \ddots & \vdots \end{bmatrix}$ $B = \begin{bmatrix} b_{11} & \dots & b_{1n} \\ b_{21} & \dots & b_{2n} \\ \vdots & \ddots & \vdots \end{bmatrix} \in G$

s.t. $A + B = E$, $E = \begin{bmatrix} 0 & \dots & 0 \\ 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} \in G \rightarrow \text{already proven}$

$\therefore \begin{bmatrix} a_{11} + b_{11} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & \dots & a_{2n} + b_{2n} \\ \vdots & \ddots & \vdots \end{bmatrix} = \begin{bmatrix} 0 & \dots & 0 \\ 0 & \dots & 0 \\ \vdots & \ddots & \vdots \end{bmatrix}_{n \times n}$ for i, j^{th} element
 $a_{ij} + b_{ij} = 0$
 $\Rightarrow b_{ij} = -a_{ij} \forall i, j \in \mathbb{R}$

$B = \begin{bmatrix} -a_{11} & \dots & -a_{1n} \\ -a_{21} & \dots & -a_{2n} \\ \vdots & \ddots & \vdots \end{bmatrix} \Rightarrow B \in G$ (as all it's elements are real)
Hence INVERSE exists.

Thus $(G, +)$ is a group when $G =$ set of all $n \times n$ real matrices.

Ring :- A ring is an algebraic structure $(R, +, *)$ which satisfies the following :-

① $(R, +)$ is an Abelian group

- a) $(a+b)+c = a+(b+c) \quad \forall a, b, c \in R$
- b) $\forall a \in R \exists e \in R$ s.t. $a+e = e+a = a$
- c) $\forall a \in R \exists b \in R$ s.t. $a+b = b+a = e$
- d) $\forall a, b \in R, a+b = b+a$

② $(R, *)$ is a Monoid :- (If commutative then R is a field)

- a) $(a*b)*c = a*(b*c) \quad \forall a, b, c \in R$
- b) Identity : $a*e' = e'*a = a \quad [\forall a \in R, \exists \text{ such } e' \in R]$

③ Distributivity of $*$ over $+$: $(a+b)*c = (a*c) + (b*c)$
 $a*(b+c) = (a*b) + (a*c)$

Not included in this definition of a ring

a) multiplicative inverse

b) Commutativity of Multiplication

↳ if (b) is included, we get a field, but M.I is still unsure

Note:

① $a \otimes (b \otimes c) = (a \otimes b) \otimes c \rightarrow$ Semi-group

↓

$e \in G \rightarrow$ Monoid

↓

$a^{-1} \rightarrow$ Group

$a \otimes b = b \otimes a$

↓
Abelian Group

② Sub-group : $(H, \otimes) \subset (G, \otimes)$

$$G \otimes G \rightarrow G$$

$$H \otimes H \rightarrow H$$

Simply taking a subset of the group's set does not give a sub-group.
 Identity and Inverse may not be in H .

(Q.) Verify whether a set of polynomials with real coefficients is a ring?

$$(R, +, *) \quad , \quad R = \left\{ \sum_{i=0}^n a_i x^i \mid a_i \in \mathbb{R}, n \in \mathbb{N} \cup \{0\} \right\}$$

① Checking if $(R, +)$ is Abelian

$$\left. \begin{aligned} \text{Let } A, B, C \in \mathbb{R} \\ A &= a_0 + a_1x + \dots + a_nx^n \\ B &= b_0 + b_1x + \dots + b_nx^n \\ C &= c_0 + c_1x + \dots + c_nx^n \end{aligned} \right\} a_i, b_i, c_i \in \mathbb{R}$$

$$\text{Now } A+B = (a_0+b_0) + (a_1+b_1)x + \dots + (a_n+b_n)x^n$$

$$(A+B)+C = [(a_0+b_0)+c_0] + [(a_1+b_1)+c_1]x + \dots + [(a_n+b_n)+c_n]x^n$$

Since, associativity is known for real nos is known

$$[(a_i+b_i)+c_i]x^i = [a_i+(b_i+c_i)]x^i$$

Hence, Associativity is proven for $(\mathbb{R}, +)$ $\rightarrow$ Rest can be proven.

LINEAR VECTOR SPACES

Q.) Let V be a set of ordered pairs (a, b) , $a, b \in \mathbb{R}$.
The following operations are defined

$$(i) (a, b) \otimes (c, d) = (a+c, b+d)$$

$$(ii) k \odot (a, b) = (ka, 0), k \in \mathbb{R}$$

Identify whether V is a vector space over $\mathbb{R}$.

- An algebraic structure $(V, F, \otimes, \odot)$ equipped with a non-empty set V , field F , a binary operation $\otimes : V \times V \rightarrow V$ and external mapping $\odot : F \times V \rightarrow V$, which satisfies:

① $(V, \otimes)$ is an Abelian Group

$$\textcircled{2} a \odot (u \otimes v) = (a \odot u) \otimes (a \odot v)$$

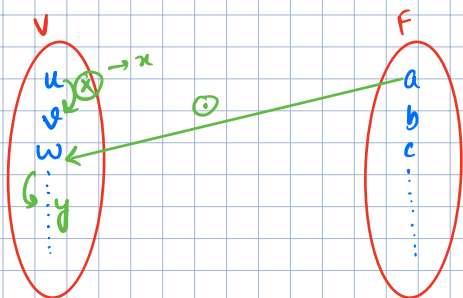
$$(u \otimes v) \odot b = (u \odot b) \otimes (v \odot b)$$

$\rightarrow$ left distributive

$u, v \in V$
 $a, b \in F$

$\downarrow$ right distributive

$\odot$ is distributive over $\otimes$



Here we consider F as an external set and not V , because:

- ① output is in V
- ② V is our set of interest

This altogether makes a SPACE.

- In a space, elements depend on other elements, contrary to a set, where elements are well defined.

Q.) $V = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, x_i \in \mathbb{R}, n \in \mathbb{Z}^+ \right\}$

$\otimes = +$ $\odot = \times$ $F = \mathbb{R}$

Is this a vector space?

Sol'n $u, v, w \in V$ s.t.

(i) $u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ $w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$ $u_i, v_i, w_i \in \mathbb{R}$

$$(u+v)+w = \begin{bmatrix} u_1+v_1 \\ u_2+v_2 \\ \vdots \\ u_n+v_n \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} u_1+v_1+w_1 \\ u_2+v_2+w_2 \\ \vdots \\ u_n+v_n+w_n \end{bmatrix}$$

$$u+(v+w) = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1+w_1 \\ v_2+w_2 \\ \vdots \\ v_n+w_n \end{bmatrix} = \begin{bmatrix} u_1+v_1+w_1 \\ u_2+v_2+w_2 \\ \vdots \\ u_n+v_n+w_n \end{bmatrix}$$

Hence, $(u_i+v_i)+w_i = u_i+(v_i+w_i) \rightarrow$ Associativity of Real Numbers

(ii) $u_i + e_i = u_i$

$\nexists e_i = 0$ where $e_i \in e = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$
 $\downarrow$ Identity element exists

(iii) Inverse exists (try yourself)

(iv) $a \times (u_i + v_i)$ $a \in F, u_i, v_i \in V$
 $\downarrow \quad \downarrow \quad \downarrow$
 $\in \mathbb{R} \quad \in \mathbb{R} \quad \in \mathbb{R}$

$= (a \times u_i) + (a \times v_i)$ Distributivity

Similar proof for right distributivity.

$\rightarrow$ Now if this exists, then the elements of set V is called a vector
 $\hookrightarrow$ ordered n -tuple
 $\downarrow$
 dimensions

→ Notation: vector V, matrix M

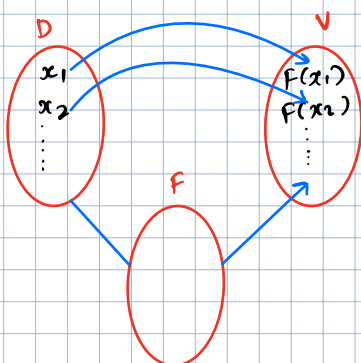
→ Hence if we say "vector spaces" we mean to say a set of vectors, which need not necessarily mean column matrices.

(Q.) $F(x) = 2x$
 $g(x) = \sin(x)$
 by plotting graphs, we can conclude linearity.

$y = x^2 + c$
 $dy/dx = 2x \rightarrow$ linear, if we don't solve
 $dy/dx = y^2 \sin(x)$

Now, let $F(x_1) = 2x_1$, $F(x_2) = 2x_2$
 $F(x_1 + x_2) = 2x_1 + 2x_2$
 $= F(x_1) + F(x_2)$

$$F(cx_1) = c F(x_1)$$



Def'n of Linearity

Now,

Let $\frac{dy}{dx} = 2y$ $\rightarrow$ Linear without solving?
 $\hookrightarrow$ NON-linear after you solve? ($y = ce^{2x}$)

$$\frac{d}{dx}(y_1 + y_2) = 2(y_1 + y_2) = \frac{dy_1}{dx} + \frac{dy_2}{dx}$$

$$\frac{d}{dx}(cy_1) = 2(cy_1) = c(2y_1) = c \frac{dy_1}{dx}$$

LINEAR

∴ linearity is defined for OPERATORS and NOT MEMBERS OF SET.

Q.) let V be a set of ordered pairs (a, b) , $a, b \in \mathbb{R}$.
 The following operations are defined

$$(i) (a, b) \otimes (c, d) = (a+c, b+d)$$

$$(ii) k \odot (a, b) = (ka, b), k \in \mathbb{R}$$

Identify whether V is a vector space over $\mathbb{R}$.

coming back to the original question

firstly, $V \equiv$ set of ordered pairs (a, b)
 $F \equiv \mathbb{R}$
 $\otimes$ } defined.

$V, \otimes \rightarrow$ Abelian group

① distributive over $\otimes$

let $u = (a_1, b_1)$
 $v = (a_2, b_2)$
 $w = (a_3, b_3) \quad \forall a, b \in \mathbb{R}$

① To check if $(V, \otimes)$ is an Abelian Group

(i) $(v \otimes v) \otimes w$

$$= \begin{pmatrix} a_1 + a_2 \\ b_1 + b_2 \end{pmatrix} \otimes w = \begin{pmatrix} a_1 + a_2 + a_3 \\ b_1 + b_2 + b_3 \end{pmatrix}$$

Associative

$$u \otimes (v \otimes w) \leftarrow \begin{pmatrix} a_1 + (a_2 + a_3) \\ b_1 + (b_2 + b_3) \end{pmatrix}$$

(ii) Identity element

$$e = (e_1, e_2)$$

$$e \otimes u = u \otimes e = u$$

$$e = (0, 0) \rightarrow \text{Identity exists}$$

(iii) Inverse element

$$u \otimes v = e$$

$$(a_1 + a_2, b_1 + b_2) = (0, 0)$$

$$a_2 = -a_1, \quad b_2 = -b_1 \Rightarrow v = (-a_1, -b_1) \rightarrow \text{Inverse exists}$$

(iv) Commutative

$$u \otimes v = (a_1 + a_2, b_1 + b_2)$$

$$= (a_2 + a_1, b_2 + b_1) = v \otimes u \rightarrow \text{COMMUTATIVE}$$

② ① is distributive over $\otimes$

$$k \odot (u \otimes v) = (k \odot u) \otimes (k \odot v)$$

$$k \odot (a_1 + a_2, b_1 + b_2) = k \odot (a_1, b_1) \otimes k \odot (a_2, b_2)$$

$$(k(a_1 + a_2), 0) = (ka_1, 0) \otimes (ka_2, 0)$$

$$(k(a_1 + a_2), 0) = (k(a_1 + a_2), 0)$$

$$\text{LHS} = \text{RHS}$$

V is a vector space over R

(Q) Let S be the set of all elements of the form

$$(x+2y, y, -x+3y) \in \mathbb{R}^3 \quad x, y \in \mathbb{R}$$

Prove that S is a subspace of $\mathbb{R}^3$

$$\text{let } u = (x_1 + 2y_1, y_1, -x_1 + 3y_1) \quad x_1, y_1 \in \mathbb{R}$$

$$v = (x_2 + 2y_2, y_2, -x_2 + 3y_2) \quad x_2, y_2 \in \mathbb{R}$$

$$\textcircled{1} \quad u+v \in S \Rightarrow (x_1+x_2 + 2(y_1+y_2), y_1+y_2, -(x_1+x_2) + 3(y_1+y_2))$$

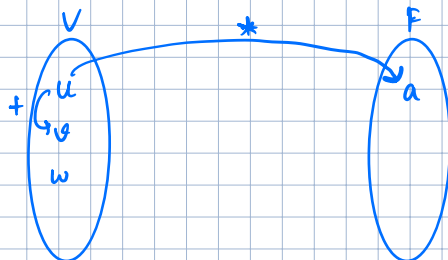
$\downarrow$ $x_3 \in \mathbb{R}$ $\downarrow$ $y_3 \in \mathbb{R} \Rightarrow u+v \in S$

$$\textcircled{2} \quad cu \in S \Rightarrow (cx_1 + 2cy_1, cy_1, -cx_1 + 3cy_1)$$

$\downarrow$ $x_4 \in \mathbb{R}$ $\downarrow$ $y_4 \in \mathbb{R} \Rightarrow cu \in S$

Associativity/Commutativity of real no.'s.

LINEAR COMBINATION OPERATION



$$\begin{aligned} + &: V \times V \rightarrow V \\ * &: F \times V \rightarrow V \end{aligned}$$

$$\underline{w} = a \underline{u} + b \underline{v}$$

linear combination

(Q) Express the polynomial $F(x)$ as a linear combination of ϕ_i 's

$$F(x) = x^2 + 4x - 3$$

$$\phi_1(x) = x^2 - 2x + 5$$

$$\phi_2(x) = 2x^2 - 3x$$

$$\phi_3(x) = x + 3$$

$$\text{let } F(x) = a \phi_1(x) + b \phi_2(x) + c \phi_3(x) \quad (a, b, c \in F)$$

comparing coefficients

$$a + 2b = 1$$

$$-2a - 3b + c = 4$$

$$5a + 3c = -3$$

$$\Rightarrow \begin{pmatrix} 1 & 2 & 0 \\ -2 & -3 & 1 \\ 5 & 0 & 3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ -3 \end{pmatrix}$$

$$a = -3, b = 2, c = 4$$

(Q.) Consider $\underline{v}_1 = [1, 2, 3]^T$, $\underline{v}_2 = [2, 3, 1]^T$, $\underline{v} = [a, b, c]^T$. Find the conditions on a, b, c so that $\underline{v}$ is a linear combination of $\underline{v}_1$ & $\underline{v}_2$.

$$\underline{v} = p\underline{v}_1 + q\underline{v}_2 \quad p, q \in F$$

So basically solution should exist

$$\text{So, } \begin{pmatrix} a \\ b \\ c \end{pmatrix} = p \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + q \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

Augmented matrix

$$\left(\begin{array}{cc|c} 1 & 2 & a \\ 2 & 3 & b \\ 3 & 1 & c \end{array} \right)$$

$$R_2' \rightarrow R_2 - 2R_1$$

$$R_3' \rightarrow R_3 - 3R_1$$

$$\left(\begin{array}{cc|c} 1 & 2 & a \\ 0 & -1 & b-2a \\ 0 & -5 & c-3a \end{array} \right)$$

$$R_3' \rightarrow R_3 - 5R_2$$

$$\left(\begin{array}{cc|c} 1 & 2 & a \\ 0 & -1 & b-2a \\ 0 & 0 & 7a-5b+c \end{array} \right)$$

$$\rightarrow = 0$$

(Q.) Show that the vectors $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ span $\mathbb{R}^2$ over $\mathbb{R}$

sol'n let $a, b \in \mathbb{R}$

$$v = av_1 + bv_2 \quad (\text{i.e. of vectors in question})$$

$$v = \begin{pmatrix} a \\ a \end{pmatrix} + \begin{pmatrix} b \\ 2b \end{pmatrix} = \begin{pmatrix} a+b \\ a+2b \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix} \text{ (say),}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix}$$

Augmented

$$\left(\begin{array}{cc|c} 1 & 1 & p \\ 1 & 2 & q \end{array} \right) \quad R_2 \rightarrow R_2 - R_1$$

$$\left(\begin{array}{cc|c} 1 & 1 & p \\ 0 & 1 & q-p \end{array} \right) \quad R_1 \rightarrow R_1 - R_2$$

$$\left(\begin{array}{cc|c} 1 & 0 & 2p-q \\ 0 & 1 & q-p \end{array} \right) \quad \begin{array}{l} a = 2p-q \\ b = q-p \end{array}$$

$\forall a, b \in \mathbb{R} \quad \forall p, q \in \mathbb{R} \Rightarrow v_1, v_2 \text{ span } \mathbb{R}^2 \text{ over } \mathbb{R}$
(i.e. any $v \in \mathbb{R}^2$)

($2p-q \in \mathbb{R}$ and $q-p \in \mathbb{R}$, $\because p, q \in \mathbb{R}$)

LINEAR INDEPENDENCE, BASIS, DIMENSION

(Q.) $f(x) = 2x^3 + x^2 + x + 1$

$$g(x) = x^3 + 3x^2 + x - 2$$

$$h(x) = x^3 + 2x^2 - x + 3$$

$$a f(x) + b g(x) + c h(x) = 0$$

$$x^3(2a+b+c) + x^2(a+3b+2c) + x(a+b-c) + (a-2b+3c) = 0 \quad \rightarrow \textcircled{1}$$

$$\text{If } \textcircled{1} \Rightarrow a=0, b=0, c=0$$

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 1 & 3 & 2 & 0 \\ 1 & 1 & -1 & 0 \end{array} \right)$$

$$R_2' \rightarrow R_2 - R_1/2, \quad R_3' \rightarrow R_3 - R_1/2, \quad R_4' \rightarrow R_4 - \frac{R_1}{2}$$

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & 5/2 & 3/2 & 0 \\ 0 & 1/2 & -3/2 & 0 \\ 0 & -5/2 & 5/2 & 0 \end{array} \right) \quad \begin{array}{l} R_2' \rightarrow 2R_2 \\ R_3' \rightarrow 2R_3 \end{array} \quad R_4' \rightarrow -2/5 R_4$$

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & 5 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right) \quad R_4' \rightarrow R_4 - R_1$$

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & 5 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -2 & 0 \end{array} \right) \quad \begin{array}{l} -2c = 0 \Rightarrow c = 0 \\ b - 3c = 0 \Rightarrow b = 3c \Rightarrow b = 0 \\ \Rightarrow \boxed{a = 0, b = 0, c = 0} \rightarrow \text{Linear Independence} \end{array}$$

Q.1) Check linear independence of $f(x), g(x)$

$$f(x) = x, \quad g(x) = e^{2x}$$

$$\text{Let, } af(x) + bg(x) = 0$$

$$ax + be^{2x} = 0$$

Wronskian Determinant (if $= 0$, l.d else l.i)

$$W = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = \begin{vmatrix} x & e^{2x} \\ 1 & 2e^{2x} \end{vmatrix} = (2x-1)e^{2x} \neq 0 \text{ in general}$$

Hence f & g are linearly independent.

$$\text{If } h(x) = \sin x$$

Oct 20

$$|W| = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix} = \begin{vmatrix} x & e^{2x} & \sin x \\ 1 & 2e^{2x} & \cos x \\ 0 & 4e^{2x} & -\sin x \end{vmatrix}$$

$$= e^{2x} [5 \sin x - 2x (\sin x + 2 \cos x)] \neq 0 \text{ in general} \\ \therefore \underline{\underline{L.I}}$$

(Q.) Determine whether $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \end{bmatrix}$, $v_3 = \begin{bmatrix} 2 \\ 5 \\ 6 \\ 4 \end{bmatrix}$, $v_4 = \begin{bmatrix} 2 \\ 6 \\ 8 \\ 5 \end{bmatrix}$

forms a basis of $\mathbb{R}^4$ over $\mathbb{R}$.

If not determine the dimension of the subspace they span.

Sol'n

$$av_1 + bv_2 + cv_3 + dv_4 = 0$$

a, b, c are L.I. $\rightarrow$

$$\left[\begin{array}{cccc|c} 1 & 1 & 2 & 2 & 0 \\ 0 & 1 & 3 & 4 & 0 \\ 0 & 0 & -2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$d = -a, \quad c = -a$$

$$b = -a, \quad a = a$$

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = a \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$$

Similarly,

$$(x_i, y_i, z_i) = x_i (1, 0, 0) + y_i (0, 1, 0) + z_i (0, 0, 1)$$

↳ Real, so field is $\mathbb{R}$

Any random vector (x_i, y_i, z_i) can be written as a linear combination of $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$.

↓
L.I. also

∴ $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ are linearly independent & span the whole space ⇒ ∴ BASIS

Basis is a linearly independent set of vectors which span the whole vector space

CARDINALITY OF BASIS = DIMENSION OF THE VECTOR SPACE

(*) NULL SPACE & RANGE SPACE

$$\begin{aligned} x + y &= 2 \\ x + 2y + z &= 4 \\ y + 2z + t &= 4 \\ z + 2t &= 3 \end{aligned}$$

Augmenting with those vectors

$$\left[\begin{array}{cccc|ccc} 1 & 1 & 0 & 0 & 0 & 2 & b_1 \\ 1 & 2 & 1 & 0 & 0 & 4 & b_2 \\ 0 & 1 & 2 & 1 & 0 & 4 & b_3 \\ 0 & 0 & 1 & 2 & 0 & 3 & b_4 \end{array} \right] \rightarrow \left[\begin{array}{cccc|ccc} 1 & 1 & 0 & 0 & 0 & 2 & b_1 \\ 0 & 1 & 1 & 0 & 0 & 2 & b_2 - b_1 \\ 0 & 1 & 2 & 1 & 0 & 4 & b_3 \\ 0 & 0 & 1 & 2 & 0 & 3 & b_4 \end{array} \right]$$

$$\left[\begin{array}{cccc|ccc} \boxed{1} & 1 & 0 & 0 & 0 & 2 & b_1 \\ 0 & \boxed{1} & 1 & 0 & 0 & 2 & b_2 - b_1 \\ 0 & 0 & \boxed{1} & 1 & 0 & 2 & b_3 - b_2 + 1 \\ 0 & 0 & 0 & \boxed{1} & 0 & 1 & b_4 - b_3 + b_2 - b_1 \end{array} \right] \leftarrow \left[\begin{array}{cccc|ccc} 1 & 1 & 0 & 0 & 0 & 2 & b_1 \\ 0 & 1 & 1 & 0 & 0 & 2 & b_2 - b_1 \\ 0 & 0 & 1 & 1 & 0 & 2 & b_3 - b_2 + 1 \\ 0 & 0 & 1 & 2 & 0 & 3 & b_4 \end{array} \right]$$

Rank = 4

If we wanted to solve $\underline{A} \underline{x} = \underline{0}$ (homogeneous), we'll do a simple back substitution

$$\left. \begin{aligned} t &= 0 \\ z + t &= 0 \Rightarrow z = 0 \\ y + z &= 0 \Rightarrow y = 0 \end{aligned} \right\} (0, 0, 0, 0) \text{ is the only sol'n for } \underline{A} \underline{x} = \underline{0} \cdot \text{ Hence } \underline{A} \underline{x} = \underline{b} \text{ will}$$

$x+y=0 \Rightarrow x=0$ ✓ have a unique solution.

$$\underline{A} \underline{x} = \underline{b}$$

$$\begin{aligned} t &= 1 \\ z+t &= 2 \rightarrow z=1 \\ y+z &= 2 \rightarrow y=1 \\ x+y &= 2 \rightarrow x=1 \end{aligned}$$

Now, for $\underline{A} \underline{x} = \underline{b}$, does the sol'n exist for all $\underline{b}$?

Will we always get a solution?

(Q.) $\Rightarrow$

$$\begin{aligned} x_1 + x_2 + x_3 &= 1 \\ 2x_1 + 3x_2 + x_3 &= 2 \\ 5x_1 + 6x_2 + 4x_3 &= 1 \end{aligned} \Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 2 \\ 5 & 6 & 4 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & -1 & 0 \\ 0 & 2 & 0 & -3 \end{array} \right] \quad (\text{Directly No Solution})$$

Now solving it properly,

$$\left[\begin{array}{ccc|c|c|c} 1 & 1 & 1 & 0 & 1 & b_1 \\ 2 & 3 & 1 & 0 & 2 & b_2 \\ 5 & 6 & 4 & 0 & 1 & b_3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c|c|c} 1 & 1 & 1 & 0 & 1 & b_1 \\ 0 & 1 & -1 & 0 & 0 & b_2 - 2b_1 \\ 0 & 1 & -1 & 0 & -4 & b_3 - 5b_1 \end{array} \right]$$

rank = 2

$$\left[\begin{array}{ccc|c|c|c} \boxed{1} & 1 & 1 & 0 & 1 & b_1 \\ 0 & \boxed{1} & -1 & 0 & 0 & b_2 - 2b_1 \\ 0 & 0 & \boxed{0} & 0 & -4 & b_3 - b_2 - 3b_1 \end{array} \right]$$

Now, for $\underline{A} \underline{x} = \underline{0}$

$$\begin{aligned} x_2 - x_3 &= 0 \\ x_1 + x_2 + x_3 &= 0 \end{aligned}$$

# eq'n = 2	let	$x_2 = \alpha$	$\Rightarrow$	$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \alpha \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$
# var = 3		$x_3 = \alpha$		
DOF = 1		$x_1 = -2\alpha$		

You have non-zero, infinitely many solutions for $\underline{A} \underline{x} = \underline{0}$. You can generate infinite solutions by scalar multiplication of α .

$\Rightarrow$ All solutions belong to the linear vector space of $(-2, 1, 1)$

↓ dimensions = 1 ↓ basis

This space is called "NULL SPACE"

When $\text{rank}(A) < n$, ∞ solutions

→ Condition for Solubility : $b_3 - b_2 - 3b_1 = 0$

rank(A) = 2

rank(A|b) should be 2.

eq's = 1

variable = 3

DoF = 2

, Let $b_1 = \beta$, $b_2 = \gamma$, $b_3 = 3\beta + \gamma$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} \beta \\ \gamma \\ 3\beta + \gamma \end{bmatrix} = \beta \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

dimension = 2

This vector space, for vector b, which renders your system of equations solvable $\Rightarrow$ Range Space

Solutions will never be unique.

(Q.)

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 2 \\ 4 & -1 & 9 \\ 2 & 1 & 1 \end{bmatrix}$$

$$b = [3 \ 0 \ 0 \ 2]^T$$

4 eq's, 3 variables

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 2 & 0 & 3 & b_1 \\ 2 & 1 & 2 & 0 & 0 & b_2 \\ 4 & -1 & 9 & 0 & 0 & b_3 \\ 2 & 1 & 1 & 0 & 2 & b_4 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 2 & 0 & 3 & b_1 \\ 1 & 2 & 0 & 0 & -3 & b_2 - b_1 \\ -5 & 3.5 & 0 & 0 & -13.5 & b_3 - 4.5b_1 \\ 1.5 & 1.5 & 0 & 0 & 0.5 & b_4 - 0.5b_1 \end{array} \right]$$

$$\downarrow$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 2 & 0 & 3 & b_1 \\ 1 & 2 & 0 & 0 & -3 & b_2 - b_1 \\ -6.75 & 0 & 0 & 0 & -8.25 & b_3 - 4.5b_1 - 1.75b_2 \\ 0.75 & 0 & 0 & 0 & 2.75 & b_4 - 0.5b_1 - 0.75b_2 \end{array} \right]$$

Clearly for $\underline{Ax} = \underline{0} \Rightarrow x = [0, 0, 0]^T$

$$\left. \begin{array}{l} b_3 - 4.5b_1 - 1.75b_2 \\ b_4 - 0.5b_1 - 0.75b_2 \end{array} \right\} = 0 \quad b_3 + b_4 - 5b_1 - 3.5b_2 = 0$$

$$\Rightarrow 3b_4 + b_3 - 4b_2 - 2b_1 = 0$$

$$\begin{aligned} \text{var} &= 4 \\ \text{eq'n} &= 1 \quad \Rightarrow \text{DOF} = 3 \end{aligned}$$

$$\text{Let } b_1 = \alpha, \quad b_2 = \beta, \quad b_4 = \gamma \quad \Rightarrow \quad b_3 = 2\alpha + 2\beta - 3\gamma$$

$$\Rightarrow \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

$\downarrow$ Range $\downarrow$ Space
 $\text{Dim} = 3$

If b can written as a L.C. of the range space $\Rightarrow$ solvable

DRAW
CHART
HERE !!

LINEAR TRANSFORMATION

$$v_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$a \in F$$

$$v_3 = a \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$v_4 = a \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + b \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \times \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$$

→ Linear transformation maps two new different linear vector spaces

$$t: \mathbb{R}^{2 \times 1} \rightarrow \mathbb{R}^{3 \times 1}$$

→ A set of polynomial & differential operator → operator
(Algebraic)

→ $S = \{ a x : a \in \mathbb{R}, a > 0 \}$ & differential operator

Transformation
→

$$S = \{ a : a \in \mathbb{R}, a > 0 \}$$

Linearity → $\underline{v}_1, \underline{v}_2, \alpha$

$$\left. \begin{array}{l} \textcircled{1} \quad t(\underline{v}_1 + \underline{v}_2) = t(\underline{v}_1) + t(\underline{v}_2) \\ \textcircled{2} \quad t(\alpha \underline{v}_1) = \alpha t(\underline{v}_1) \end{array} \right\} \quad t(\alpha \underline{v}_1 + \beta \underline{v}_2) = \alpha t(\underline{v}_1) + \beta t(\underline{v}_2)$$

(Q) Let V be a set of continuous differential functions with the following operators defined

$$T: V \rightarrow V \quad T[F(x)] = \int_0^x F(t) dt \quad \forall F(x) \in V; x \in \mathbb{R}$$

Identify if it's linear or not?

Sol'n

$$\begin{aligned} \textcircled{1} \quad T[\int_0^x F(x) + g(x)] &= \int_0^x [F(t) + g(t)] dt = \int_0^x F(t) dt + \int_0^x g(t) dt \\ &= T[F(x)] + T[g(x)] \end{aligned}$$

$$\textcircled{2} \quad T[\alpha F(x)] = \int_0^x \alpha F(t) dt = \alpha \int_0^x F(t) dt = \alpha T[F(x)]$$

→ LINEAR

Q) Identify if they are linear or not

$$t: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$a) \quad t(x, y, z) = (x + 2y - 3z, 4x - 5y + 6z)$$

$$b) \quad t: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$t(x, y, z) = (x+1, y+1)$$

Aus) (a) Let $v_1 = (x_1, y_1, z_1) \in \mathbb{R}^3 \Rightarrow \alpha v_1 = (\alpha x_1, \alpha y_1, \alpha z_1)$
 $v_2 = (x_2, y_2, z_2)$
 $\alpha v_2 = (\alpha x_2, \alpha y_2, \alpha z_2)$

$$\Rightarrow t(v_1) = (x_1 + 2y_1 - 3z_1, 4x_1 - 5y_1 + 6z_1)$$

$$t(v_2) = (x_2 + 2y_2 - 3z_2, 4x_2 - 5y_2 + 6z_2)$$

$$\alpha v_1 + \beta v_2 = (\underbrace{\alpha x_1 + \beta x_2}_{x_3}, \underbrace{\alpha y_1 + \beta y_2}_{y_3}, \underbrace{\alpha z_1 + \beta z_2}_{z_3})$$

$$\begin{aligned} t(\alpha v_1 + \beta v_2) &= [\alpha x_1 + \beta x_2 + 2(\alpha y_1 + \beta y_2) - 3(\alpha z_1 + \beta z_2), \\ &\quad 4(\alpha x_1 + \beta x_2) - 5(\alpha y_1 + \beta y_2) + 6(\alpha z_1 + \beta z_2)] \\ &= (x_3 + 2y_3 - 3z_3, 4x_3 - 5y_3 + 6z_3) \end{aligned}$$

$$t(\alpha v_1) = (\alpha x_1 + 2\alpha y_1 - 3\alpha z_1, 4\alpha x_1 - 5\alpha y_1 + 6\alpha z_1)$$

$$t(\beta v_2) = (\beta x_2 + 2\beta y_2 - 3\beta z_2, 4\beta x_2 - 5\beta y_2 + 6\beta z_2)$$

$$\text{If } x_1, x_2 \in \mathbb{R} \quad \therefore \alpha x_1 + \beta x_2 \in \mathbb{R}$$

Based on above : LINEAR

b) Let $v_1 = (x_1, y_1, z_1) \in \mathbb{R}^3 \Rightarrow \alpha v_1 = (\alpha x_1, \alpha y_1, \alpha z_1)$
 $v_2 = (x_2, y_2, z_2)$
 $\beta v_2 = (\beta x_2, \beta y_2, \beta z_2)$

$$\alpha v_1 + \beta v_2 = (\alpha x_1 + \beta x_2, \alpha y_1 + \beta y_2, \alpha z_1 + \beta z_2)$$

$$t(\alpha v_1 + \beta v_2) = (\underbrace{\alpha x_1}_{x_3} + \underbrace{\beta x_2}_{y_3} + 1, \underbrace{\alpha y_1}_{z_3} + \underbrace{\beta y_2}_{z_3} + 1) \\ = (x_3 + 1, y_3 + 1)$$

$$t(\alpha v_1) = (\alpha x_1 + 1, \alpha y_1 + 1)$$

$$t(\beta v_2) = (\beta x_2 + 1, \beta y_2 + 1) \quad (\text{Not linear})$$

$$a) \quad \hat{L} = \frac{d^2}{dx^2} - m^2$$

$$b) \quad \hat{L} = \frac{\partial}{\partial t} - \alpha \frac{\partial^2}{\partial z^2} + \beta \quad \rightarrow \text{conservation}$$

Ans a) Let F_1 & F_2 be two continuous functions $\in V$

$$\hat{L}F_1 = \frac{d^2 F_1}{dx^2} - m^2 F_1 \quad \hat{L}F_2 = \frac{d^2 F_2}{dx^2} - m^2 F_2$$

$$\hat{L}(F_1 + F_2) = \frac{d^2}{dx^2} (F_1 + F_2) - m^2 (F_1 + F_2)$$

$$= \frac{d^2 F_1}{dx^2} - m^2 F_1 + \frac{d^2 F_2}{dx^2} - m^2 F_2$$

$$= \hat{L}F_1 + \hat{L}F_2$$

$$\hat{L}(\alpha F_1) = \frac{d^2}{dx^2} (\alpha F_1) - m^2 (\alpha F_1)$$

$$= \alpha \frac{d^2 F_1}{dx^2} - \alpha m^2 F_1 = \alpha \left(\frac{d^2 F_1}{dx^2} - m^2 F_1 \right)$$

$$= \alpha \hat{L}F_1 \quad (\text{LINEAR})$$

b) Let $F_1(t, z)$ and $F_2(t, z)$ be two continuous functions $\in V$

$$\hat{L}F_1 = \frac{\partial F_1}{\partial t} - \alpha \frac{\partial^2 F_1}{\partial z^2} + \beta$$

$$\hat{L}F_2 = \frac{\partial F_2}{\partial t} - \alpha \frac{\partial^2 F_2}{\partial z^2} + \beta$$

$$\hat{L}(F_1 + F_2) = \frac{\partial}{\partial t}(F_1 + F_2) - \alpha \frac{\partial^2}{\partial z^2}(F_1 + F_2) + \beta$$

$$= \left(\frac{\partial F_1}{\partial t} - \alpha \frac{\partial^2 F_1}{\partial z^2} + \beta \right) + \left(\frac{\partial F_2}{\partial t} - \alpha \frac{\partial^2 F_2}{\partial z^2} \right)$$

NOT LINEAR

(Q-) let $B = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\} \mathbb{R}^3(\mathbb{R})$

$$t: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} ; t \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} ; t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Determine $t \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$

$$\alpha \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + \gamma \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \begin{matrix} \nearrow \\ \in \mathbb{R} \end{matrix}$$

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -1 & 0 & 1 & x \\ 0 & 1 & -1 & y \\ 1 & -1 & 1 & z \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} -1 & 0 & 1 & x \\ 0 & 1 & -1 & y \\ 0 & -1 & 2 & x+y+z \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} -1 & 0 & 1 & x \\ 0 & 1 & -1 & y \\ 0 & 0 & 1 & x+y+z \end{array} \right]$$

$$\Rightarrow x+y+z = \gamma \quad \checkmark$$

$$y = \beta - \gamma$$

$$\begin{aligned} -\alpha + \gamma &= x \\ y + z &= \alpha \quad \checkmark \end{aligned}$$

$$x+2y+z = \beta \quad \checkmark$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + \gamma \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$t \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \left(\alpha \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right)$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$t \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$t \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} y+z \\ x+2y+z \\ x+y+z \end{bmatrix}$$

$$t \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

Consider a vector space $\mathbb{R}^n$. Prove that $\mathbb{R}^n$ is inner product space with inner product defined as

$$\mathbb{R}^n, \langle \underline{u}, \underline{v} \rangle = \sum_{i=1}^n u_i v_i \quad (\text{standard inner product for real inner system}).$$

Is it binary operation?

$$\mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R} \quad (\because \text{not } \mathbb{R}^n, \text{ therefore not binary operation})$$

Inner product can be defined in any way it just needs to satisfy the following conditions:-

$$\textcircled{1} \text{ Linearity} \rightarrow \langle \underline{u} + \underline{v}, \underline{w} \rangle = \langle \underline{u}, \underline{w} \rangle + \langle \underline{v}, \underline{w} \rangle$$

$$\langle \alpha \underline{u}, \underline{v} \rangle = \alpha \langle \underline{u}, \underline{v} \rangle$$

or

$$\langle \alpha \underline{u} + \beta \underline{v}, \underline{w} \rangle = \alpha \langle \underline{u}, \underline{w} \rangle + \beta \langle \underline{v}, \underline{w} \rangle$$

Proof:

let $\underline{u}, \underline{v}, \underline{w}$ be vectors in vector space $\mathbb{R}^n$.

$$\rightarrow \left. \begin{aligned} \underline{u} &= [u_1, u_2, \dots, u_n]^T \\ \underline{v} &= [v_1, v_2, \dots, v_n]^T \\ \underline{w} &= [w_1, w_2, \dots, w_n]^T \end{aligned} \right\}$$

$$\text{let } \underline{z} = \underline{u} + \underline{v} = [u_1 + v_1, u_2 + v_2, \dots, u_n + v_n]^T$$

$$\langle \underline{u} + \underline{v}, \underline{w} \rangle = \langle \underline{z}, \underline{w} \rangle = \sum z_i w_i$$

$$= \sum (u_i + v_i) w_i$$

$$= \sum u_i w_i + \sum v_i w_i$$

$$= \langle \underline{u}, \underline{w} \rangle + \langle \underline{v}, \underline{w} \rangle$$

Let, $\underline{u} = \alpha \underline{u}$

$$\underline{u} = [\alpha u_1, \alpha u_2, \dots, \alpha u_n]^T$$

$$\langle \alpha \underline{u}, \underline{v} \rangle = \sum \alpha u_i v_i = \alpha \langle \underline{u}, \underline{v} \rangle$$

(ii) Symmetry : $\langle \underline{u}, \underline{v} \rangle = \langle \underline{v}, \underline{u} \rangle$

(iii) Positive Definiteness : $\langle \underline{u}, \underline{u} \rangle \geq 0$

$$\langle \underline{u}, \underline{u} \rangle = \sum_{i=1}^n u_i^2 \geq 0 \quad (u_i^2 \geq 0; \text{ real number square is always greater than 0})$$

$= 0$ if & only if $\underline{u} = \underline{0}$!

Ex:-

$$\underline{u} = \begin{bmatrix} 2+2i \\ 5 \end{bmatrix}, \quad \underline{v} = \begin{bmatrix} 3i \\ 5 \end{bmatrix}$$

$$\langle \underline{u}, \underline{v} \rangle = \sum_{i=1}^2 u_i v_i = (2+2i)3i + 25 = 6i - 6 + 25 = 6i + 19$$

$$\langle \underline{v}, \underline{u} \rangle = \langle \underline{u}, \underline{v} \rangle = 19 + 6i$$

Ex:-

$$\underline{u} = \begin{bmatrix} 2+2i \\ 3 \end{bmatrix}, \quad \underline{v} = \begin{bmatrix} 3i \\ 5 \end{bmatrix}, \quad \underline{w} = \begin{bmatrix} 3 \\ 5i \end{bmatrix}$$

$$\langle \underline{u}, \underline{u} \rangle = (2+2i)(2-2i) + 9 = 4 + 8i - 4 + 9 = 13$$

$$\langle \underline{v}, \underline{v} \rangle = 25 - 9 = 16$$

$$\langle \underline{w}, \underline{w} \rangle = -25 + 9 = -16 \quad (\text{positive definiteness is not satisfied})$$

Since we cannot change the rules, so we need to change the formula

Standard Inner Product : $\langle \underline{u}, \underline{v} \rangle = \sum_{i=1}^n u_i \overline{v_i}$

$$\langle \underline{u}, \underline{u} \rangle = (2+2i)(2-2i) + 9 = 4 + 4 + 9 = 17$$

$$\langle \underline{v}, \underline{v} \rangle = (3i)(-3i) + 25 = 9 + 25 = 34$$

$$\langle \underline{w}, \underline{w} \rangle = 9 + 25 = 34$$

In this case the 3rd condition is always satisfied. ✓

So finally, conditions are as follows

1. Linearity : $\langle \alpha \underline{u} + \beta \underline{v}, \underline{w} \rangle = \alpha \langle \underline{u}, \underline{w} \rangle + \beta \langle \underline{v}, \underline{w} \rangle$

2. Symmetry : $\langle \underline{u}, \underline{v} \rangle = \overline{\langle \underline{v}, \underline{u} \rangle}$

3. Positive Definiteness : $\langle \underline{u}, \underline{u} \rangle \geq 0$

$$\begin{aligned}\langle \underline{u}, \alpha \underline{v} \rangle &= \overline{\langle \alpha \underline{v}, \underline{u} \rangle} = \overline{\alpha \langle \underline{v}, \underline{u} \rangle} \\ &= \overline{\alpha} \overline{\langle \underline{v}, \underline{u} \rangle} = \overline{\alpha} \langle \underline{u}, \underline{v} \rangle\end{aligned}$$

Ex:- Determine the value of λ which makes the following operation inner product -

$$\underline{x}, \underline{y} \in \mathbb{R}^2$$

$$\langle \underline{x}, \underline{y} \rangle = x_1 y_1 - 3x_1 y_2 - 3x_2 y_1 + \lambda x_2 y_2$$

for this to be inner product

$$\langle \underline{x}, \underline{x} \rangle \geq 0$$

$$\langle \underline{x}, \underline{x} \rangle = x_1^2 - 3x_1 x_2 - 3x_2 x_1 + \lambda x_2^2 \geq 0$$

$$x_1^2 - 6x_1 x_2 + \lambda x_2^2 \geq 0$$

$$\left(\frac{x_1}{x_2}\right)^2 - 6\left(\frac{x_1}{x_2}\right) + \lambda \geq 0 \quad \Rightarrow \quad y^2 - 6y + \lambda \geq 0$$

for this to be true $b^2 - 4ac \leq 0$

$$36 - 4\lambda \leq 0$$

$$4\lambda \geq 36$$

$$\lambda \geq 9$$

Ex:- Determine a non-zero vector in $\mathbb{R}^3$ which is orthogonal to $\underline{v}_1, \underline{v}_2, \underline{v}_3$.

$$\underline{v}_1 = \begin{bmatrix} 1 & 1 & 2 \end{bmatrix}^T$$

$$\underline{v}_2 = \begin{bmatrix} 2 & 1 & 3 \end{bmatrix}^T$$

$$\underline{v}_3 = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}^T$$

$$\text{let } \underline{x} = \begin{bmatrix} a & b & c \end{bmatrix}^T$$

$$\langle \underline{x}, \underline{v}_1 \rangle = 0$$

$$a + b + 2c = 0$$

$$\langle \underline{x}, \underline{v}_2 \rangle = 0$$

$$2a + b + 3c = 0$$

$$\langle \underline{x}, \underline{v}_3 \rangle = 0$$

$$a + 2b + 3c = 0$$

$$\begin{bmatrix} 1 & 1 & 2 & 0 \\ 2 & 1 & 3 & 0 \\ 1 & 2 & 3 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 2 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 2 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} a + b + 2c &= 0 \\ b + c &= 0 \end{aligned}$$

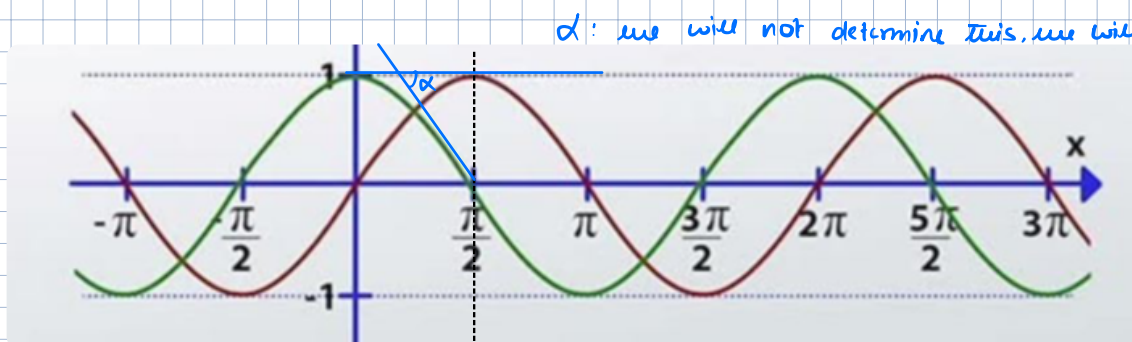
$$\text{DoF} = 3 - 2 = 1 \quad \Phi \quad c = \alpha, \quad b = -\alpha, \quad a = \alpha$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$\therefore$ we have a vector $[1, -1, 1]^T$ & every displacement is orthogonal.

(Q.) Verify the following functions are orthogonal

- $\sin x, \cos x$ in $[-\pi, \pi]$
- $\sin n\pi x, \sin m\pi x$ in $[-1, 1]$
- $\sin n\pi x, \cos m\pi x$ in $[-1, 1]$
- $\cos n\pi x, \cos m\pi x$ in $[-1, 1]$



$$\therefore \langle f(x), g(x) \rangle = \int_a^b f(x) \overline{g(x)} dx$$

a) $f(x) = \sin x, \quad g(x) = \cos x$

$$\langle f(x), g(x) \rangle = \int_{-\pi}^{\pi} \sin x \cos x dx = \frac{1}{2} \int_{-\pi}^{\pi} \sin 2x dx = 0$$

b) $f(x) = \sin n\pi x, \quad g(x) = \sin m\pi x$ in $[-1, 1]$

$$\langle f(x), g(x) \rangle = \int_{-1}^1 \sin(n\pi x) \sin(m\pi x) dx$$

$$= \frac{1}{2} \int_{-1}^1 [\cos(m-n)\pi x - \cos(m+n)\pi x] dx.$$

$$= \frac{1}{2} \left\{ \frac{\sin(m-n)\pi x}{(m-n)\pi} - \frac{\sin(m+n)\pi x}{(m+n)\pi} \right\}_{-1}^1$$

How this becomes 0?

Solve (b), (c), (d) part by yourself

(#) $f(x) = \sum_{n=1}^{\infty} a_n \sin(n\pi x)$ Fourier Series Expansion

Taking inner product with $\sin(m\pi x)$ from -1 to $+1$

$$\langle f(x), \sin(m\pi x) \rangle = \int_{-1}^1 \sum_{n=1}^{\infty} a_n \sin(n\pi x) \sin(m\pi x) dx$$

(Q.) Let B be the vector space of all the polynomial over $\mathbb{R}$ degree less than or equal to 2.

Determine a polynomial which is orthogonal to $f(t)$ if

$$\langle f, g \rangle = \int_0^1 f(t) g(t) dt \quad f(t) = 2t+1$$

Sol'n: Let $g(t) = a_0 + a_1 t + a_2 t^2$

Now determining a_0, a_1 and a_2 such that -

$$\int_0^1 (a_0 + a_1 t + a_2 t^2) (2t+1) dt = 0$$

$$\int_0^1 (2a_0 t + a_0 + 2a_1 t^2 + a_1 t + 2a_2 t^3 + a_2 t^2) dt = 0$$

$$\int_0^1 (2a_2 t^3 + (2a_1 + a_2) t^2 + (a_1 + 2a_0) t + a_0) dt = 0$$

$$\frac{2a_2}{4} + \frac{2a_1 + a_2}{3} + \frac{a_1 + 2a_0}{2} + a_0 = 0$$

$$6a_2 + 8a_1 + 4a_2 + 6a_1 + 12a_0 + 12a_0 = 0$$

$$14a_1 + 10a_2 + 24a_0 = 0$$

$$5a_2 + 7a_1 + 12a_0 = 0$$

$$\text{DOF} = 3 - 1 = 2 \Rightarrow$$

$$a_1 = \alpha, a_2 = \beta$$

$$a_0 = -\frac{5\beta}{12} - \frac{7}{12}\alpha$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \alpha \begin{bmatrix} -7/12 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -5/12 \\ 0 \\ 1 \end{bmatrix}$$

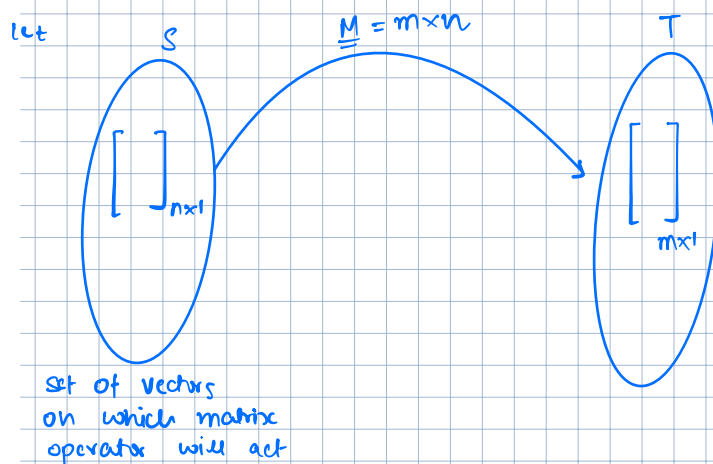
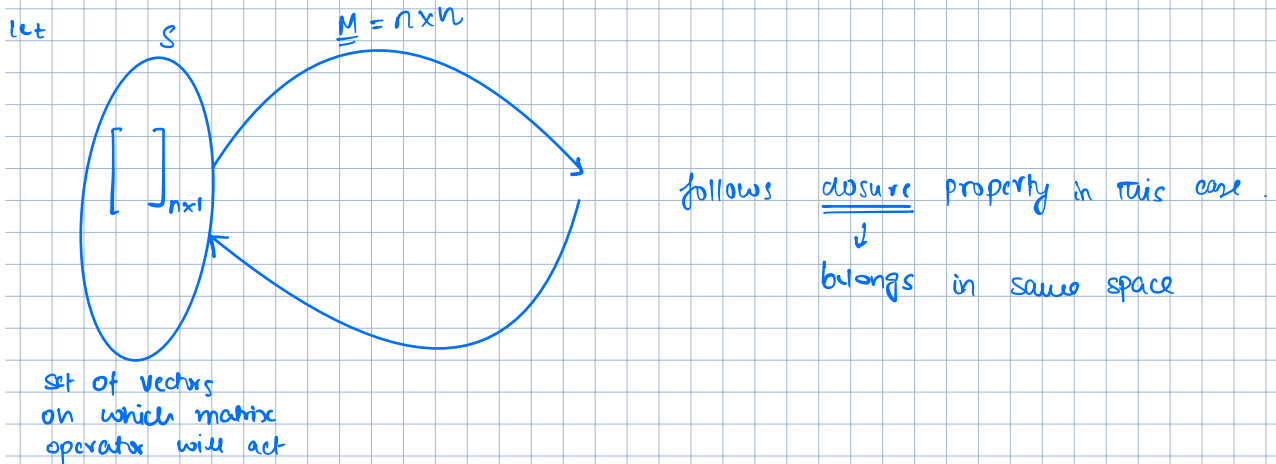
$$= \alpha_1 \begin{bmatrix} -7 \\ 12 \\ 0 \end{bmatrix} + \beta_1 \begin{bmatrix} -5 \\ 0 \\ 12 \end{bmatrix}$$

COMPLETE NOTES OF 4/10/23

Adjoint of an operator

If $L = \frac{d^2}{dx^2} + x \frac{d}{dx}$; $x = 0$, $\frac{dy}{dx} = 0$

$x = 1$, $\frac{dy}{dx} + u = 0$



(#) Imagine we have a matrix A

$$\underline{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} ; \underline{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\text{operate } \underline{A} \text{ on } \underline{u} \Rightarrow \underline{A} \underline{u} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} a_{11}u_1 + a_{12}u_2 \\ a_{21}u_1 + a_{22}u_2 \end{bmatrix}$$

→ belong to the same set $\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

$$\langle \underline{A} \underline{u}, \underline{v} \rangle = a_{11} u_1 v_1 + a_{12} u_2 v_1 + a_{21} u_1 v_2 + a_{22} u_2 v_2$$

↓
Taking inner product

$$= u_1 (a_{11} v_1 + a_{21} v_2) + u_2 (a_{12} v_1 + a_{22} v_2)$$

$$\langle \underline{A} \underline{u}, \underline{v} \rangle = \left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} a_{11} v_1 + a_{21} v_2 \\ a_{12} v_1 + a_{22} v_2 \end{bmatrix} \right\rangle$$

$$= \left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\rangle$$

$$= \langle \underline{u}, \underline{B} \underline{v} \rangle$$

↓
Adjoint of operator A.

for square matrices of order n the adjoint operator would simply be the transpose, but what about non-square matrices.

$$(\#) \quad \underline{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}, \quad \underline{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \underline{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\underline{A} \underline{u} = \begin{bmatrix} a_{11} u_1 + a_{12} u_2 + a_{13} u_3 \\ a_{21} u_1 + a_{22} u_2 + a_{23} u_3 \end{bmatrix}$$

$$\langle \underline{A} \underline{u}, \underline{v} \rangle = \text{can't be solved/proved further}$$

$$\text{Since we know, } \langle a \underline{u}, \underline{v} \rangle = a \langle \underline{u}, \underline{v} \rangle$$

$$\langle \underline{u}, a \underline{v} \rangle = \bar{a} \langle \underline{u}, \underline{v} \rangle$$

To arrive at the adjoint of a matrix operator, first take the transpose A then take it's conjugate. $\rightarrow$ most general form.

$$(\#) \quad \underline{A} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}, \quad \text{Adjoint} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} = \underline{A} \quad (\text{self-adjoint operator})$$

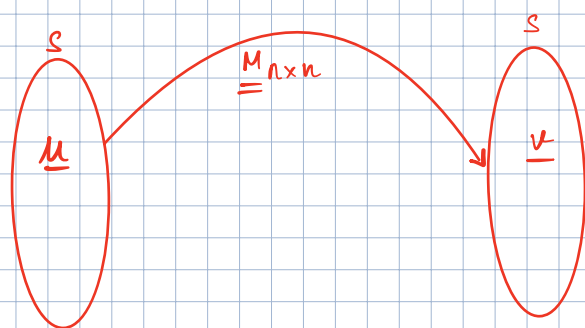
1.) The eigenvalues of self-adjoint operators are always real. ~ obviously only for square matrices.

$$\langle \underline{A} \underline{u}, \underline{u} \rangle = \langle \lambda \underline{u}, \underline{u} \rangle \quad (\because \underline{A} \underline{u} = \lambda \underline{u})$$

crossing both sides by $\underline{u}$

$\underline{u}$: eigenvector

scaling factor / eigenvalue



(λ is the eigenvalue of $\underline{A}$ for the eigenvector $\underline{u}$.)

$\downarrow$
triplet is important.

$$\begin{aligned} \langle \underline{A} \underline{u}, \underline{u} \rangle &= \langle \lambda \underline{u}, \underline{u} \rangle \\ &= \lambda \langle \underline{u}, \underline{u} \rangle \end{aligned} \quad \left| \quad \begin{aligned} \langle \underline{u}, \underline{A} \underline{u} \rangle &= \langle \underline{u}, \lambda \underline{u} \rangle \\ &= \bar{\lambda} \langle \underline{u}, \underline{u} \rangle \end{aligned} \right.$$

since $\underline{A}$ is self Adjoints, the inner products are equal $\therefore \lambda = \bar{\lambda}$
 $\downarrow$
eigenvalues are real.

2.) The eigenvectors of a self-adjoint operator corresponding to distinct eigenvalues are always orthogonal.

$$\lambda_1 \rightarrow \underline{v}_1$$

$$\langle \underline{A} \underline{v}_1, \underline{v}_2 \rangle = \lambda_1 \langle \underline{v}_1, \underline{v}_2 \rangle$$

$$\lambda_2 \rightarrow \underline{v}_2$$

$$\langle \underline{v}_1, \underline{A} \underline{v}_2 \rangle = \lambda_2 \langle \underline{v}_1, \underline{v}_2 \rangle \quad (\because \lambda_2 = \bar{\lambda}_2)$$

subtract

$$(\lambda_1 - \lambda_2) (\langle \underline{v}_1, \underline{v}_2 \rangle) = 0 \quad \Rightarrow \quad \boxed{\langle \underline{v}_1, \underline{v}_2 \rangle = 0}$$

$\downarrow$
orthogonal

Fredholm's alternative theorem

(i) Determine the range space of the following system of equations.

$$x_1 + x_2 + x_3 = 1$$

$$2x_1 + 3x_2 + x_3 = 2$$

$$5x_1 + 6x_2 + 4x_3 = 1$$

Sol'n

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 2 & 3 & 1 & b_2 \\ 5 & 6 & 4 & b_3 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & 1 & -1 & b_2 - 2b_1 \\ 0 & 1 & -1 & b_3 - 5b_1 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & 1 & -1 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_3 - b_2 - 3b_1 \end{array} \right]$$

for solvability $\text{rank}(A) = \text{rank}(A|b)$

$$b_3 - b_2 - 3b_1 = 0$$

$$\text{DOF} = 3 - 1 = 2$$

$$\Rightarrow b_1 = \alpha, \quad b_2 = \beta, \quad b_3 = \beta + 3\alpha$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

↓ always linearly independent

Fredholm's Alternative Theorem $\Rightarrow$

$\underline{A} \underline{x} = \underline{b}$ is solvable

$$\text{if } \langle \underline{b}, \underline{y} \rangle = 0$$

$$\underline{A}^* \underline{y} = 0 \quad (\underline{y} \text{ belongs to the null space of adjoint of } A)$$

$$\text{let } A = \begin{bmatrix} 1 & 2 & 5 \\ 1 & 3 & 6 \\ 1 & 1 & 4 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 1 & 3 & 6 & 0 \\ 1 & 1 & 4 & 0 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right]$$

$$x_1 + 2x_2 + 5x_3 = 0$$

$$x_2 + x_3 = 0$$

$$\text{DOF} = 1$$

$$x_2 = \alpha, \quad x_3 = -\alpha, \quad x_1 = 3\alpha$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \alpha \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} = \alpha \underline{v}, \quad \underline{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\text{if } \langle \underline{b}, \underline{y} \rangle = 0$$

$$3b_1 + b_2 - b_3 = 0$$

$$b_1 = \beta, \quad b_2 = \gamma, \quad b_3 = 3\beta + \gamma$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \beta \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$